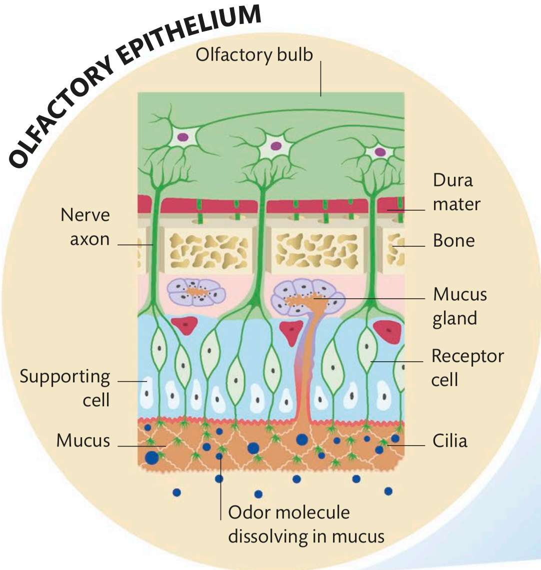

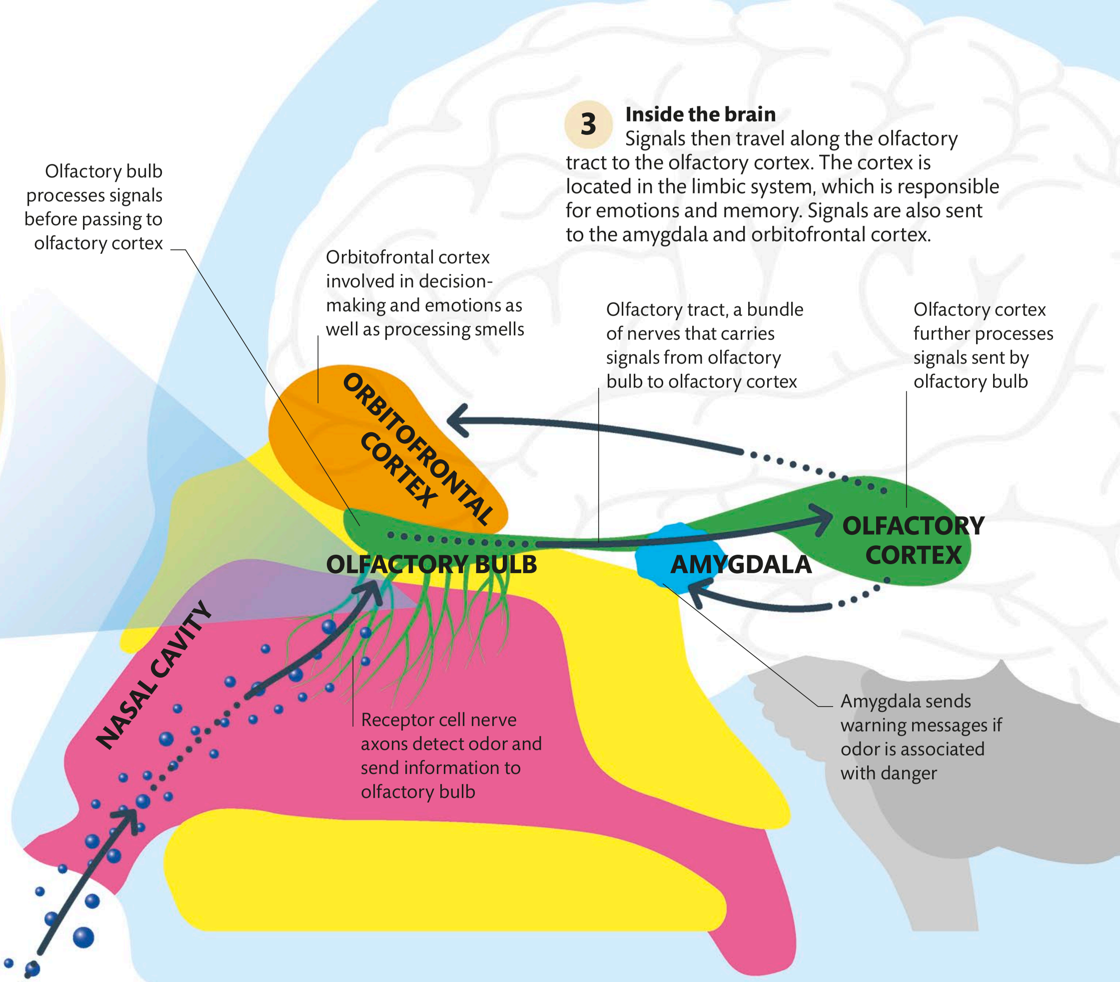

3 Inside the brain
Signals then travel along the olfactory tract to the olfactory cortex. The cortex is located in the limbic system, which is responsible for emotions and memory. Signals are also sent to the amygdala and orbitofrontal cortex.

2 Olfactory receptors
Each odor molecule activates a particular combination of olfactory receptors. The activated receptor cells send impulses up through nerve axons to the olfactory bulb for processing.

1 Smell enters the nose
Odor molecules are drawn up through the nose and warmed to enhance the scent. The molecules dissolve in mucus produced by the olfactory epithelium and stimulate cilia that are connected to receptor cells.

12 MILLION
THE NUMBER OF **OLFACTORY CELLS** IN THE **HUMAN BODY**

Capturing a scent

When we inhale, odor molecules drift into the nose and activate receptor cells in the nasal cavity, triggering a reflex to breathe in more deeply. In the nasal cavity, the odors dissolve in the mucus that covers a sheet of neurons and supporting cells called olfactory epithelium. The molecules spread through the mucus to hairlike structures called cilia that are attached to receptor cells. These cells send signals to the olfactory bulb—a structure located in the forebrain that forms part of the brain's limbic system. Data is then sent to various parts of the brain, particularly the olfactory cortex.